



Maven Market - DATA ANALYSIS



Company Overview

Maven Market is a mid-sized retail business operating a network of stores across multiple regions, serving a diverse customer base through a broad range of products and brands. The business focuses on managing store performance, product sales, customer relationships, and day-to-day retail operations across its locations.

Analysing Maven Market's 1997-1998 business data offers an opportunity to understand how different stores, products, customers, regions, and returns contribute to overall business performance and where operational improvements may be needed.



Objective of the Analysis

This analysis evaluates Maven Market's sales and operational performance by identifying trends and differences across stores, products, customers, and regions. The analysis aims to uncover key business opportunities and operational issues that can support better

decisions around store efficiency, product performance, customer behavior, and returns.



Content



Sales & Store Performance Report



Product Performance Report



Returns & Operational Performance Report



Customer Performance Report



Sales & Store Performance Report

How do store types rank based on total revenue generated across 1997–1998?

▼ Result

- **Business Goal:** To identify which store types are driving the most revenue and determine where the business should focus its resources and growth efforts.

```
-- How do store types rank based on total revenue generated across 1997-1998?
```

```
WITH all_tran AS(
SELECT *
FROM "transaction-1997"
UNION ALL
SELECT *
FROM "transaction-1998"
),

pro_sto AS(
SELECT *
FROM all_tran t
INNER JOIN products p
ON t.product_id = p.product_id
INNER JOIN stores s
ON s.store_id = t.store_id
),

revenue_col AS(
SELECT store_type,
SUM(quantity*product_retail_price) AS total_revenue
FROM pro_sto
GROUP BY store_type)

SELECT *,
ROW_NUMBER() OVER(ORDER BY total_revenue DESC) AS store_rank
FROM revenue_col
```

Output Messages Notifications

SQL

Showing rows: 1 to 5 Page No: 1

store_type	total_revenue	store_rank
Supermarket	789601.26	1
Deluxe Supermarket	669726.65	2
Gourmet Supermarket	152980.27	3
Mid-Size Grocery	118534.14	4
Small Grocery	33704.12	5

- **Insight:** Supermarkets generated the highest total revenue at \$789K, ranking #1, while Small Grocery stores generated the lowest at \$34K, ranking #5. This shows a substantial revenue gap between store types, with larger-format stores driving the majority of overall revenue

How does store performance differ across sales districts and regions?

▼ Result

- **Business Goal:** To evaluate how store performance varies across sales districts and regions, so management can identify high-

performing markets, address underperforming areas, and make informed decisions about resource allocation and regional growth.

```
Query  Query History
1  -- How does store performance differ across sales districts and regions?
2
3  WITH all_tran AS(
4  SELECT *
5  FROM "transaction-1997"
6  UNION ALL
7  SELECT *
8  FROM "transaction-1998"
9  ),
10
11  sto_pro AS(
12  SELECT *
13  FROM all_tran t
14  INNER JOIN products p
15  ON p.product_id = t.product_id
16  INNER JOIN stores s
17  ON s.store_id = t.store_id
18  ),
19
20  reg AS(
21  SELECT *
22  FROM sto_pro
23  INNER JOIN regions r
24  ON r.region_id = sto_pro.region_id
25  ),
26
27  revenue_col AS(
28  SELECT sales_district,
29  SUM(quantity*product_retail_price) AS total_revenue
30  FROM reg
31  GROUP BY sales_district
32  )
33
34  SELECT *,
35  ROW_NUMBER() OVER(ORDER BY total_revenue DESC) AS district_rank
36  FROM revenue_col
37  ORDER BY total_revenue DESC
38
39
```

- **Insight:** Los Angeles is the top-performing sales district with approximately \$209K in revenue, while Guadalajara ranks lowest at about \$5K. The large revenue gap indicates significant variation in district performance and highlights districts that may require closer investigation.

Query	Query History
1	-- How does store performance differ across sales districts and regions?
2	
3	WITH all_tran AS(
4	SELECT *
5	FROM "transaction-1997"
6	UNION ALL
7	SELECT *
8	FROM "transaction-1998"
9	),
10	
11	sto_pro AS(
12	SELECT *
13	FROM all_tran t
14	INNER JOIN products p
15	ON p.product_id = t.product_id
16	INNER JOIN stores s
17	ON s.store_id = t.store_id
18	),
19	
20	reg AS(
21	SELECT *
22	FROM sto_pro
23	INNER JOIN regions r
24	ON r.region_id = sto_pro.region_id
25	),
26	
27	revenue_col AS(
28	SELECT sales_region,
29	SUM(quantity*product_retail_price) AS total_revenue
30	FROM reg
31	GROUP BY sales_region
32	)
33	
34	SELECT *,
35	ROW_NUMBER() OVER(ORDER BY total_revenue DESC) AS region_rank
36	FROM revenue_col
37	ORDER BY total_revenue DESC
38	
39	

- **Insight:** North West is the strongest-performing region, generating approximately \$848K in revenue, while Central West ranks last with only about \$9K. The substantial revenue gap highlights significant regional differences in store performance and indicates opportunities to investigate and improve weaker regions.

How do stores' monthly sales volumes change over time, and which months show growth, decline, or no change for each store?

▼ Result

- **Business Goal:** To track month-to-month sales volume across stores, identify periods of growth or decline, and determine which stores show consistent changes in demand so management can improve sales planning, inventory management, and store-level performance.

```
Query    Query History
1  --How do stores' monthly sales volumes change over time, and which months show growth, decline, or no change for each store?
2
3  WITH all_tran AS(
4  SELECT transaction_date, store_id, quantity
5  FROM "transaction-1997"
6  UNION ALL
7  SELECT transaction_date, store_id, quantity
8  FROM "transaction-1998"
9  ),
10
11  month_sale AS(
12  SELECT store_id, DATE_TRUNC('month', transaction_date) AS sale_month,
13  SUM(quantity) AS total_quantity
14  FROM all_tran
15  GROUP BY store_id, DATE_TRUNC('month', transaction_date)
16  ),
17
18  monthly_change AS(
19  SELECT store_id, sale_month, total_quantity,
20  LAG(total_quantity) OVER(PARTITION BY store_id ORDER BY sale_month) AS prev_month_qty
21  FROM month_sale
22  ),
23
24  change AS(
25  SELECT *,
26  (total_quantity - prev_month_qty) AS qty_change
27  FROM monthly_change
28  )
29
30  SELECT *,
31  (CASE
32  WHEN qty_change > 0 THEN 'Growth'
33  WHEN qty_change < 0 THEN 'Decline'
34  ELSE 'No change'
35  END) AS status
36  FROM change
37
38
```

- **Insight:** Sales volumes fluctuate considerably across stores and months. Growth was strongest in November 1998, while April and October 1998 recorded the highest number of declining stores. Overall, the results indicate that store sales are highly variable and require closer monitoring for seasonal or store-specific patterns.



Product Performance Report

Which are the top 10 product brands by total revenue across 1997–1998?

▼ Result

- **Business Goal:** Identify the highest-revenue product brands to understand which brands contribute most to the company's sales and where management should focus merchandising, inventory, and promotional efforts.

```
--Which are the top 10 product brands by total revenue across 1997-1998?
```

```
WITH all_tran AS(  
  SELECT *  
  FROM "transaction-1997"  
  UNION ALL  
  SELECT *  
  FROM "transaction-1998"  
)  
  
pro AS(  
  SELECT *  
  FROM all_tran t  
  INNER JOIN products p  
  ON p.product_id = t.product_id  
)  
  
SELECT product_brand,  
SUM(product_retail_price*quantity) AS total_revenue  
FROM pro  
GROUP BY product_brand  
ORDER BY SUM(product_retail_price*quantity) DESC  
LIMIT 10
```

- **Insight:** Hermanos is the top-performing product brand, generating approximately \$56.7K in revenue, followed by Tell Tale at \$51.6K and Ebony at \$49.7K. The top 10 brands collectively generate about \$461K, with Hermanos leading the group by a clear margin.

Which products have the highest sales volume while generating relatively low profit per unit?

▼ Result

- **Business Goal:** Identify high-volume products that generate relatively low profit per unit, helping management evaluate whether pricing, product costs, or margins should be optimized for these products.

```
--Which products have the highest sales volume while generating relatively low profit per unit?
```

```
WITH tran_all AS(
SELECT *
FROM "transaction-1997"
UNION ALL
SELECT *
FROM "transaction-1998"
),

pro AS(
SELECT *
FROM tran_all t
INNER JOIN products p
ON p.product_id = t.product_id
)

SELECT product_name,
SUM(quantity) AS total_qty,
(product_retail_price - product_cost) AS profit_per_unit
FROM pro
GROUP BY product_name, product_retail_price, product_cost
ORDER BY SUM(quantity) DESC, (product_retail_price - product_cost) ASC
LIMIT 10
```

- **Insight:** Tell Tale Fresh Lima Beans has the highest sales volume at 698 units but generates a relatively low profit per unit of \$1.93. Steady Whitening Toothpaste also has a high sales volume at 684 units but only earns \$0.34 profit per unit, indicating strong demand but limited profitability.

Which products generate revenue above or below their brand's average, and how can they be classified as "Above Average" or "Below Average" revenue performers?

▼ Result

- **Business Goal:** Identify products that outperform or underperform their brand's average revenue to help the business recognise strong product performers and identify products that may require further attention or improvement.

```

--Which products generate revenue above or below their brand's average, and how can they be classified as "Above Average" or "Below Average" revenue performers

WITH all_tran AS(
SELECT +
FROM "transaction-1997"
UNION ALL
SELECT +
FROM "transaction-1998"
),

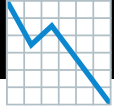
pro AS(
SELECT t.product_id, product_name, product_brand,
SUM(quantity*product_retail_price) AS total_revenue
FROM all_tran t
INNER JOIN products p
ON p.product_id = t.product_id
GROUP BY t.product_id, product_name, product_brand
),

brand_avg AS(
SELECT +,
AVG(total_revenue) OVER(PARTITION BY product_brand) AS brand_avg_revenue
FROM pro
)

SELECT product_id, product_name, product_brand, total_revenue, ROUND((brand_avg_revenue),2) AS brand_avg,
(CASE
WHEN total_revenue > brand_avg_revenue THEN 'above average'
WHEN total_revenue < brand_avg_revenue THEN 'below average'
ELSE 'Equal to average'
END) AS revenue_status
FROM brand_avg

```

- **Insight:** Out of 1,559 products, 756 (48.5%) generate revenue above their brand average, while 798 (51.2%) fall below average and only 5 are exactly at the average. This shows that product performance is fairly evenly split, with slightly more products underperforming their respective brand benchmarks.



Returns & Operational Performance Report

Which stores have the highest return rates, after accounting for their sales volume?

▼ Result

- **Business Goal:** Identify stores experiencing unusually high returns rather than simply flagging items with a small number of returns.

Query	Query History
1	--Which stores have the highest return rates, after accounting for their sales volume?
2	
3	WITH all_tran AS(
4	SELECT *
5	FROM "transaction-1997"
6	UNION ALL
7	SELECT *
8	FROM "transaction-1998"
9	),
10	
11	total_sales AS(
12	SELECT t.store_id, store_name,
13	SUM(quantity) AS sold_qty
14	FROM all_tran t
15	INNER JOIN stores s
16	ON s.store_id = t.store_id
17	GROUP BY t.store_id, store_name
18	),
19	
20	return_qty AS(
21	SELECT s.store_id, store_name,
22	SUM(quantity) AS returned_qty
23	FROM returns r
24	INNER JOIN stores s
25	ON s.store_id = r.store_id
26	GROUP BY s.store_id, store_name
27	),
28	
29	main AS(
30	SELECT s.store_id, sold_qty, returned_qty
31	FROM total_sales s
32	INNER JOIN return_qty r
33	ON s.store_id = r.store_id)
34	
35	SELECT *,
36	ROUND((returned_qty :: numeric/sold_qty)*100,2) AS return_rate
37	FROM main
38	ORDER BY return_rate DESC

- **Insight:** Store 8 has the highest return rate at 1.17%, followed by Store 20 at 1.14% and Store 23 at 1.12%. These rates remain high even after accounting for sales volume, indicating that these stores may require further investigation into product or operational issues contributing to returns.

Which high-performing products have the highest return rates?

▼ Result

- **Business Goal:** To identify high-performing products that also have elevated return rates, so the business can investigate potential product-quality, customer-experience, or operational issues without overlooking their strong sales contribution.

```
Query Query History
1 --Which high-performing products have the highest return rates?
2
3 WITH all_tran AS(
4   SELECT * FROM "transaction-1997"
5   UNION ALL
6   SELECT * FROM "transaction-1998"
7 ),
8
9 pro AS(
10  SELECT t.product_id, product_name, SUM(quantity*product_retail_price) AS total_revenue
11  FROM all_tran t
12  INNER JOIN products p
13  ON t.product_id = p.product_id
14  GROUP BY t.product_id, product_name
15 ),
16
17 --Shortlisting high performing product (total_revenue > avg_revenue = "HIGH PERFORMING")
18 avg_revenue_v AS(
19  SELECT ROUND(AVG(total_revenue),2) AS avg_revenue
20  FROM(
21    SELECT t.product_id,SUM(quantity*product_retail_price) AS total_revenue
22    FROM all_tran t
23    INNER JOIN products p
24    ON p.product_id = t.product_id
25    GROUP BY t.product_id
26  ),
27
28 r_benchmark AS(
29  SELECT *
30  FROM(
31    SELECT * ,
32    (CASE
33     WHEN total_revenue > avg_revenue THEN 'above_avg'
34     WHEN total_revenue < avg_revenue THEN 'below_avg'
35     ELSE 'No change'
36    END) AS status
37  FROM pro
38  CROSS JOIN avg_revenue_v)
39  WHERE status = 'above_avg'
40 ),
41
42 pro_ret AS(
43  SELECT r.product_id, product_name, SUM(quantity*product_retail_price) AS returned_revenue
44  FROM returns r
45  INNER JOIN products p
46  ON p.product_id = r.product_id
47  GROUP BY r.product_id, product_name
48 )
49
50 SELECT b.product_id, b.product_name, total_revenue, returned_revenue,
51  ROUND((returned_revenue :: numeric/ total_revenue)*100,2) AS return_rate
52  FROM r_benchmark b
53  INNER JOIN pro_ret r
54  ON b.product_id = r.product_id
55  GROUP BY b.product_id, b.product_name, total_revenue, returned_revenue
56  ORDER BY return_rate DESC
57
```

- **Insight:** The analysis shows that several high-performing products also have relatively high return rates, meaning strong sales do not necessarily indicate a problem-free product. These products should be prioritized for further investigation to determine whether

returns are linked to product quality, customer expectations, or other operational factors.

Which product brands have the highest return rates, and are these return patterns consistent across different stores?

▼ Result

- **Business Goal:** To identify product brands with unusually high return rates and determine whether these return patterns are consistent across stores, so management can prioritize brands and locations for quality, operational, or customer-experience improvements.

```
Query Query History
1  --Which product brands have the highest return rates, and are these return patterns consistent across different stores?
2
3  WITH all_tran AS(
4    SELECT * FROM "transaction-1997"
5    UNION ALL
6    SELECT * FROM "transaction-1998"
7  ),
8
9  pro AS(
10   SELECT *
11   FROM all_tran t
12   INNER JOIN products p
13   ON p.product_id = t.product_id
14   ),
15
16  total_rev AS(
17   SELECT product_brand, SUM(quantity*product_retail_price) AS sale_revenue
18   FROM pro
19   GROUP BY product_brand
20   ),
21
22  pro_ret AS(
23   SELECT *
24   FROM products p
25   INNER JOIN returns r
26   ON r.product_id = p.product_id
27   ),
28
29  ret_revenue AS(
30   SELECT product_brand, SUM(quantity*product_retail_price) AS returned_revenue
31   FROM pro_ret
32   GROUP BY product_brand),
33
34  rate AS(
35   SELECT t.product_brand, sale_revenue, returned_revenue,
36   ROUND((returned_revenue::numeric/sale_revenue)*100,2) AS return_rate
37   FROM total_rev t
38   INNER JOIN ret_revenue r
39   ON t.product_brand = r.product_brand
40   GROUP BY t.product_brand, returned_revenue, sale_revenue
41   )
42
43  SELECT * FROM rate
44  ORDER BY return_rate DESC
```

- Insight:** King has the highest brand return rate at 1.78%, followed by Genteel 1.57% and Robust 1.44% . The variation in return rates highlights brands that may require closer investigation for potential product- or quality-related issues. Further analysis will be conducted to determine whether these return patterns are consistent across different stores or vary by location.

```

1  --Which product brands have the highest return rates, and are these return patterns consistent across different stores?
2
3  WITH all_tran AS(
4    SELECT * FROM "transaction-1997"
5    UNION ALL
6    SELECT * FROM "transaction-1998"
7  ),
8
9  prs AS(
10   SELECT l.product_id, product_brand, product_retail_price, quantity, store_id
11   FROM all_tran t
12   INNER JOIN products p
13   ON p.product_id = t.product_id
14   ),
15
16  sto AS(
17   SELECT p.store_id, product_brand, product_retail_price, quantity
18   FROM prs p
19   INNER JOIN stores s
20   ON p.store_id = s.store_id
21   ),
22
23  revenue_col AS(
24   SELECT store_id, product_brand ,
25   SUM(quantity*product_retail_price) AS sale_revenue
26   FROM sto
27   GROUP BY store_id, product_brand),
28
29  ret AS(
30   SELECT r.product_id, product_brand, quantity, product_retail_price, store_id
31   FROM returns r
32   INNER JOIN products p
33   ON p.product_id = r.product_id
34   ),
35
36  ret_sto AS(
37   SELECT r.store_id, product_brand,
38   SUM(quantity*product_retail_price) AS returned_revenue
39   FROM ret r
40   INNER JOIN stores s
41   ON r.store_id = s.store_id
42   GROUP BY r.store_id, product_brand
43   ),
44
45  sale_ret AS(
46   SELECT s.store_id, r.product_brand, sale_revenue, returned_revenue
47   FROM revenue_col s
48   INNER JOIN ret_sto r
49   ON s.store_id = r.store_id
50   AND r.product_brand = s.product_brand
51   ),
52
53  rate AS(
54   SELECT *,
55   ROUND((returned_revenue::numeric/ sale_revenue)*100,2) AS return_rate
56   FROM sale_ret
57   )
58
59  SELECT * FROM rate
60  ORDER BY return_rate DESC
61

```

- Insight:** King has the highest overall return rate at 1.78%, followed by Genteel 1.57%) and Robust 1.44% . However, return rates vary considerably across stores—for example, King ranges from 3.13% to 10.00%, while Genteel ranges from 0.89% to 7.34%. This indicates that high return rates are not fully consistent across stores and may be influenced by store-specific factors.



Customer Performance Report

How does purchasing volume differ across membership card types, and how do they rank based on quantity sold?

▼ Result

Business Goal: To compare purchasing volume across membership card types and rank them by quantity sold, so the business can understand which customer segments drive the most purchase volume and optimize membership strategies accordingly.

```
Query Query History
1  --How does purchasing volume differ across membership card types, and how do they rank based on quantity sold?
2
3  WITH all_tran AS(
4    SELECT * FROM "transaction-1997"
5    UNION ALL
6    SELECT * FROM "transaction-1998"
7  ),
8
9  cust AS(
10   SELECT member_card, SUM(quantity) AS total_qty_sold
11   FROM all_tran t
12   INNER JOIN customers c
13   ON c.customer_id = t.customer_id
14   GROUP BY member_card
15  )
16
17  SELECT
18  ROW_NUMBER() OVER(ORDER BY total_qty_sold DESC) AS membership_rank, *
19  FROM cust
```

Insight: Bronze card members have the highest purchasing volume, with 465,090 units sold, ranking #1. Normal members rank second with 192,265 units, followed by Golden (106,251) and Silver (69,883). Bronze members account for the majority of total quantity sold, indicating that they are the largest contributor to purchasing volume. However, a high quantity sold does not necessarily translate into high revenue, as revenue also depends on the price and product mix purchased by each membership group.

How does customer purchasing behaviour vary across different education and occupation groups?

▼ Result

- **Business Goal:** To understand how purchasing volume varies across education and occupation groups, so the business can identify its most valuable customer segments and tailor marketing, product offerings, and customer strategies accordingly.

```
Query  Query History
1  --How does customer purchasing behaviour vary across different education and occupation groups?
2
3  WITH all_tran AS(
4    SELECT * FROM "transaction-1997"
5    UNION ALL
6    SELECT * FROM "transaction-1998"
7  ),
8
9  cust AS(
10   SELECT education, SUM(quantity) AS total_qty
11   FROM customers c
12   INNER JOIN all_tran t
13   ON t.customer_id = c.customer_id
14   GROUP BY education
15  )
16
17
18  SELECT
19    ROW_NUMBER() OVER(ORDER BY total_qty DESC) AS education_rank,
20    *
21  FROM cust
```

- **Insight:** Customers with Partial High School education have the highest purchasing volume at 251,699 units, followed closely by High School Degree customers at 246,560 units. Graduate Degree customers have the lowest volume at 44,377 units. This indicates that purchasing volume is concentrated among customers with lower to mid-level education backgrounds.

```

Query  Query History
1  --How does customer purchasing behaviour vary across different education and occupation groups?
2
3  WITH all_tran AS(
4  SELECT * FROM "transaction-1997"
5  UNION ALL
6  SELECT * FROM "transaction-1998"
7  ),
8
9  cust AS(
10 SELECT education, SUM(quantity) AS total_qty
11 FROM customers c
12 INNER JOIN all_tran t
13 ON t.customer_id = c.customer_id
14 GROUP BY education
15 )
16
17
18 SELECT
19 ROW_NUMBER() OVER(ORDER BY total_qty DESC) AS education_rank,
20 *
21 FROM cust

```

- **Insight:** Professional customers generate the highest purchasing volume at 271,571 units, followed by Skilled Manual (224,670) and Manual (202,753) occupations. Clerical customers have the lowest volume at 14,455 units, showing a significant difference in purchasing volume across occupation groups.

Which customers have made purchases from multiple product brands, and how can customers be classified based on the diversity of products they purchase?

▼ Result

- **Business Goal:** To identify customers based on the diversity of product brands they purchase, so the business can understand customer purchasing breadth, segment customers by brand diversity, and develop targeted cross-selling and retention strategies.

```
Query Query History
1 --Which customers have made purchases from multiple product brands,
2 --and how can customers be classified based on the diversity of products they purchase?
3
4 WITH all_tran AS(
5   SELECT * FROM "transaction-1997"
6   UNION ALL
7   SELECT * FROM "transaction-1998"
8 ),
9
10 pro AS(
11   SELECT t.product_id, product_brand, customer_id
12   FROM all_tran t
13   INNER JOIN products p
14   ON p.product_id = t.product_id),
15
16 cust AS(
17   SELECT CONCAT(first_name,' ', last_name) AS full_name,
18   COUNT(DISTINCT(product_brand)) AS num_of_brands
19   FROM pro p
20   INNER JOIN customers c
21   ON c.customer_id = p.customer_id
22   GROUP BY CONCAT(first_name,' ', last_name)
23 ),
24
25 short_list AS(
26   SELECT *
27   FROM cust
28   WHERE num_of_brands > 1
29 ),
30
31 --customer diversity classification based on observed distribution of num of brands column: [(x >= 50 "High"),(20 <= x < 50 "Medium"),(x < 20 "Low")]
32
33 status AS(
34   SELECT *,
35   (CASE
36     WHEN num_of_brands >= 50 THEN 'High Diversity'
37     WHEN num_of_brands BETWEEN 20 AND 49 THEN 'Medium Diversity'
38     ELSE 'Low Diversity'
39   END) AS brand_diversity_status
40   FROM short_list
41 )
42
43 SELECT * FROM status
```

- **Insight:** The analysis shows that most customers have low brand diversity: 5,148 customers are classified as Low Diversity, compared with 2,914 Medium Diversity and 590 High Diversity customers. The number of brands purchased ranges from 2 to 83, with an average of about 20 brands per customer. This indicates that while most customers purchase from a relatively limited range of brands, a smaller group demonstrates much broader purchasing diversity.